

Hyundai

SmartReserve

An AI-Powered In-Cabin EV Charging Co-Pilot that finds free chargers in your district, predicts congestion, and physically locks a 30-minute exclusive slot all from the car screen.

Category 2: AI-Powered & Digital In-Vehicle Experience

Rs.10 Lakh Prize Pool

OCPP 1.6 Protocol

3

SOFTWARE
SCREENS

934

REAL STATION
NODES

78%

GNN
ACCURACY

30s

DEMO
TIME

Why Indian EV Drivers Are Suffering

India has over 60,000 registered EVs in Telangana alone — but the charging experience is completely broken.

P1 NO LIVE STATUS

Apps like ChargeZone and Statiq only show whether chargers exist — not whether a specific connector port is free right now. The "Available" badge is often 24 hours old.

P2 PHONE WHILE DRIVING

Drivers must pull out their phone while driving, open 3 different apps, and manually search — which is both dangerous and illegal in India.

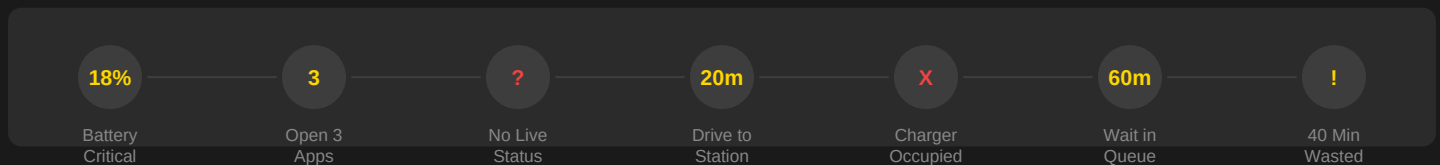
P3 NO RESERVATION

Even if you know a charger is free, someone else can arrive before you and steal your spot. There is no way to lock a connector in advance.

P4 ICE VEHICLE BLOCKING

Non-electric cars park at charging stations (called ICE-ing). There is no enforcement or economic penalty to stop this from happening.

CURRENT PAINFUL JOURNEY (WITHOUT SmartReserve)



Hyundai SmartReserve: What We Built

THE BIG IDEA — ONE SENTENCE

A software system that shows all charging station live statuses on the Hyundai in-car touchscreen, predicts waiting time using AI, and lets the driver press one button to physically lock a charger for 30 minutes — only unlockable with a secret PIN or QR code.

1

In-Cabin Dashboard

The Hyundai car touchscreen view. Shows live map, AI predictions, and the Reserve button. Driver never touches their phone.

2

Cloud AI Brain

FastAPI backend + Stanford ROLAND GNN model. Predicts congestion, manages reservations, and sends lock commands via OCPP 1.6.

3

Charger Kiosk Simulator

Simulated charger touchscreen that physically locks out unauthorized users until the correct PIN is entered.

NEW SMOOTH JOURNEY (WITH SmartReserve)

18%

Battery
Detected

AI

Best Charger
Found

1

One Tap
Reserve

LOCK

Charger
Locked

PIN

PIN
Received

18m

Drive
There

KEY

Type PIN

60kW

Charging
Starts!

WHY THIS PERFECTLY FITS IDEATHON CATEGORY 2

Category 2 asks: "How can AI, digital content, and software create new experiences inside the vehicle?" Our answer: AI predicts which charger will be free. The car screen shows this to the driver. The driver never leaves the car experience — no phone, no confusion,

How Everything Connects

END-TO-END REAL-TIME PIPELINE: 100% PURE SOFTWARE — ZERO HARDWARE

1

IN-CABIN DASHBOARD

HYUNDAI AVNT / CCNC

Battery Telematics

18% / 41km

District Station Map

Live Pins

AI Congestion Badge

GNN Queue Time

1-Touch 30-Min Lock

Reserve Slot

2

CLOUD AI BRAIN

PYTHON FASTAPI + WEBSOCKET

Stanford ROLAND GNN

R2=80.02%

OCPP 1.6 Protocol Broker

Official Spec

WebSocket Hub

<100ms Sync

Telangana Station DB

934 CSV

3

CHARGER KIOSK SIM

DIGITAL TWIN LOCKOUT SCREEN

Default State

AVAILABLE

Hardware Lockout State

RESERVED

PIN Keypad / QR Sensor

PIN: 482910

Charging Output

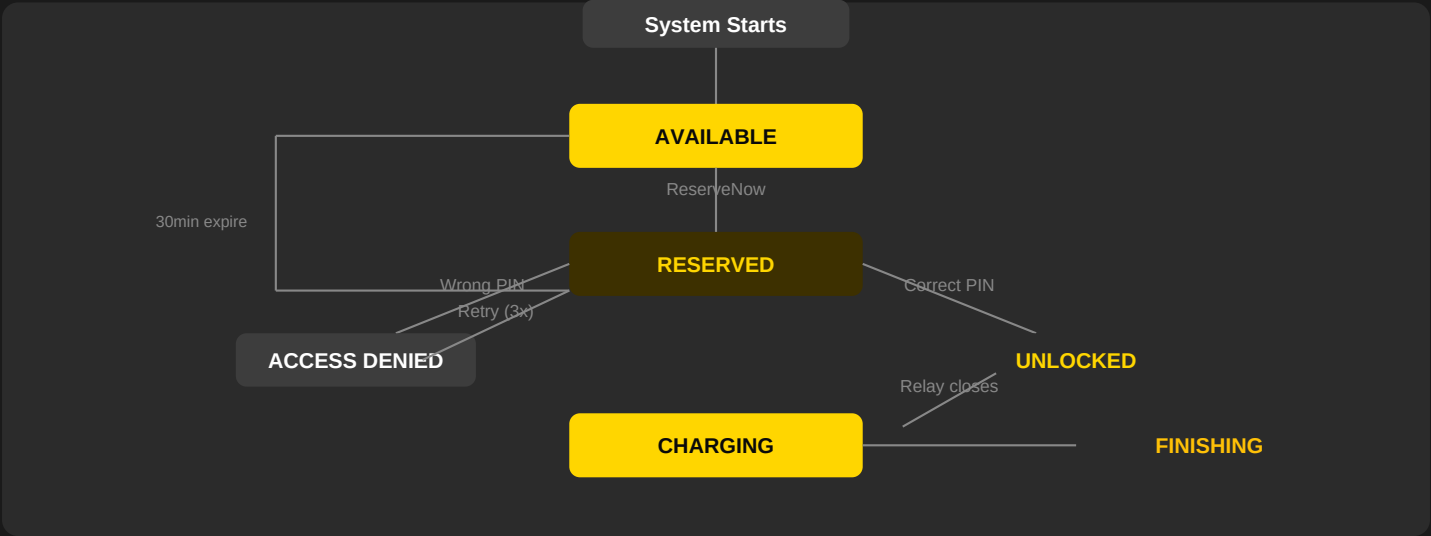
60kW DC Flow

WHAT DATA FLOWS BETWEEN THE PARTS

FROM	TO	WHAT IS SENT	PROTOCOL
Car Dashboard	Cloud API	Reserve request: Station ID, Duration 30	HTTP POST
Cloud API	Charger Kiosk	Lock command: ReserveNow with OTP token	WebSocket
Cloud API	Car Dashboard	Confirmation + 6-digit PIN + countdown t	HTTP Response
Charger Kiosk	Cloud API	PIN entered by driver for verification	HTTP POST
Cloud API	Charger Kiosk	Unlock signal: close relay, start chargi	WebSocket
Charger Kiosk	Car Dashboard	Live meter: kW, kWh, SoC% (simulated bat	WebSocket

How the 30-Minute Lock Works

CHARGER STATE MACHINE — ALL POSSIBLE STATES



WHAT THE CHARGER SCREEN SHOWS IN EACH STATE

STATE	LED COLOR	SCREEN MESSAGE	POWER STATUS
AVAILABLE	Green Solid	60 kW CCS2 Available — Plug in or tap RFID	Relay Open · Ready
RESERVED	Blue Solid	RESERVED for Hyundai Driver — Enter 6-digit P	Relay LOCKED · 0 kW
ACCESS DENIED	Red Flash	WRONG PIN — Access Denied. Connector stays lo	Relay LOCKED · 0 kW
UNLOCKED	Green Blink	Identity Verified! Plug in your cable now	Awaiting plug-in
CHARGING	Green Solid	Charging: 60 kW · SoC: 42% · 28 min remaining	Relay CLOSED · 60 kW
FINISHING	Amber Solid	Charging complete at 80%. Please unplug now.	Idle fee starts 5 min

HOW PHYSICAL LOCK WORKS (NO HARDWARE NEEDED TO DEMO)

In the real world, the charger's internal microcontroller runs the OCPP state machine. The contactor relay opens and closes based on OCPP commands. In our software simulator, this is simply a JavaScript boolean variable: relayOpen = true/false. When false, the screen

Step-by-Step Demo Flow

01

Driver in Car — Battery at 18%

System detects low SoC. Automatically queries AI for nearest free 60kW charger. Displays map on car screen.

02

GET /stations — Load 934 stations

FastAPI fetches all Telangana stations. GNN model scores each for congestion. Map shows color-coded pins.

03

Driver taps Reserve for 30 Minutes

One tap on car screen. POST /reserve sends station_id, duration:30, user_id to cloud. No phone needed.

04

Cloud generates PIN + sends ReserveNow

Backend creates 6-digit PIN (e.g. 482910). OCPP ReserveNow command pushed via WebSocket to kiosk.

05

Kiosk screen turns BLUE — LOCKED

Charger display changes from Available to Reserved. Relay stays OPEN. Any plug-in attempt is blocked.

06

Car screen shows PIN + countdown

Dashboard displays: PIN: 482910 | Slot expires in 29:58. Driver drives 1.8km to Gachibowli.

07

Unauthorized person types 111111

Kiosk flashes RED: Access Denied. Contactor remains locked. 0 kW. Proves physical lockout is real.

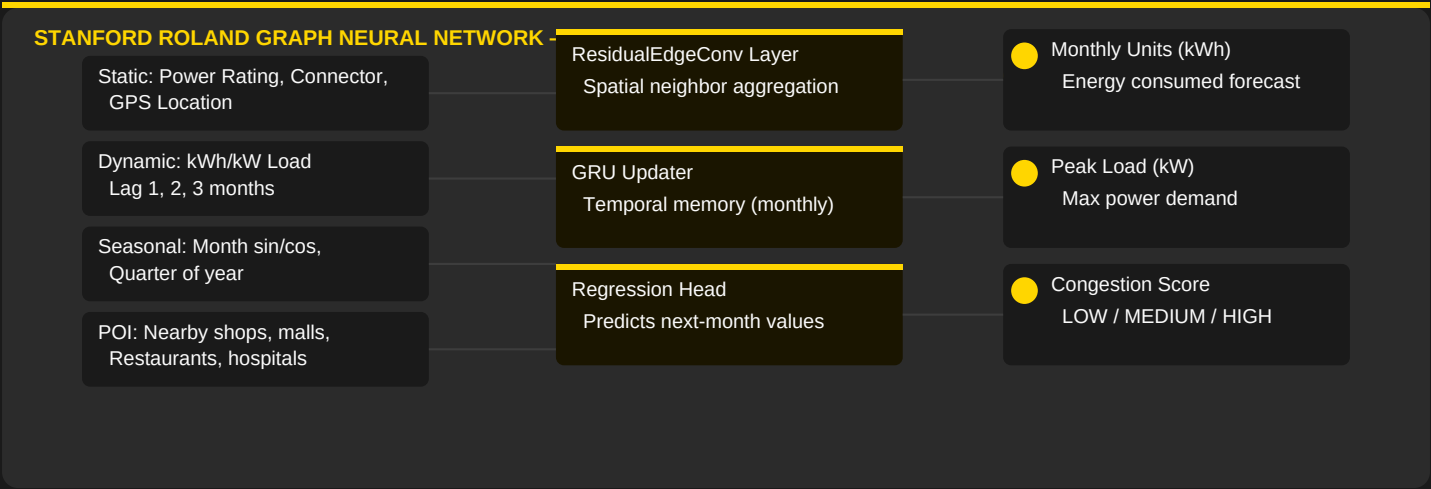
08

Driver types correct PIN 482910

Kiosk verifies via POST /verify-otp. WebSocket UNLOCK_RELAY sent. Relay CLOSSES. 60kW DC charging begins.

How the GNN Predicts Congestion

Most apps just show you whether a charger is currently free. Our AI goes further — it predicts which stations will be congested in the next few hours across the entire district, so you always pick the smartest charger, not just the nearest one.



TRAINED MODEL

GNN trained on 36 months of real data (Jan 2023 – Dec 2025) across 934 charging stations in Telangana.
Pre-trained weights: roland_final_weights_80_02.pt

REAL DATASET

Data from two official Indian government sources:
Bureau of Energy Efficiency (BEE)
Parivahan Sewa (Ministry of Road Transport). Not dummy data.

MODEL PERFORMANCE

R2 Score = 78.50% to 80.02% on test set.
Correctly explains 80% of demand variation across all 33 districts of Telangana.

COVERAGE

All 33 districts of Telangana:
Hyderabad, Rangareddy, Medchal-Malkajgiri, Sangareddy — all major Hyundai EV markets.

Technology Stack and File Structure

PROJECT FILE STRUCTURE

```

EV_Hyundai/
  EVCS_Demand_Forecasting/      (DONE)
    final_model/
      roland_final_weights_80_02.pt
    processed/nodes_master.csv
  smartreserve_backend/
    main.py                     FastAPI server, all endpoints
    websocket_manager.py        Real-time connection hub
    ocpp_engine.py              OCPP 1.6 protocol commands
    gnn_predictor.py            Loads GNN + predictions
    stations_db.py              Reads nodes_master.csv
  smartreserve_frontend/
    car_dashboard/index.html    In-Cabin UI
    kiosk_simulator/index.html  Charger UI
  run_demo.py                  One script starts everything

```

TECHNOLOGY STACK

FastAPI (Python)

Main server. API endpoints + WebSocket connections. Write a function — it becomes an API automatically. Port

Leaflet.js (Map)

Free open-source map library. Plots 934 station pins on Hyderabad street map with color-coded status. No API

WebSockets (python-websockets)

Instant two-way communication. Car reserves → kiosk screen updates in under 100 milliseconds. No page

Tailwind CSS

Makes UI look professional and Hyundai-styled (dark mode, modern automotive feel) without weeks of

PyTorch + PyTorch Geometric

Loads pre-trained Stanford ROLAND GNN. Runs inference to predict which stations will be congested.

Browser WebSocket API

Car dashboard and kiosk connect to backend WebSocket hub using built-in new WebSocket(). No library needed.

Pandas + SQLite

Reads 934-station nodes_master.csv. Stores reservations, PIN codes, session records during the demo.

OCPP 1.6 Protocol

Standard used by Tata Power and ChargeZone. ReserveNow, CancelReservation,

How to Build Everything in 7 Days

DAY 1

3-4 hrs

Set Up Backend Skeleton

Install FastAPI, Uvicorn. Create main.py with 4 endpoints: GET /stations, POST /reserve, POST /verify-otp, WS /ws/kiosk. Load nodes_master.csv into memory

DAY 2

4-5 hrs

Build the Charger Kiosk Simulator

Create kiosk/index.html. Design the charger screen with four states (Available, Reserved, Access Denied, Charging). Add PIN number keypad. Connect

DAY 3

5-6 hrs

Build the Car Dashboard

Create dashboard/index.html. Add Leaflet.js map with Hyderabad coordinates. Plot all 934 stations from the database. Add battery SoC indicator, station

DAY 4

4-5 hrs

Connect the Reservation Flow End-to-End

Wire up full journey: Click Reserve → Backend generates PIN → Backend sends WebSocket to Kiosk → Kiosk goes Blue/Locked → Dashboard shows PIN → Driver

DAY 5

4-5 hrs

Add GNN Congestion Predictions

Integrate gnn_predictor.py that loads pre-trained roland_final_weights_80_02.pt model. Run inference for current month. Convert

DAY 6

4-5 hrs

Add Live Charging Graph + Polish UI

After unlock, simulate charging session: SoC starts at 18% and increases every 2 seconds on both screens. Plot taper curve (fast to 80%, then slower). Add

DAY 7

Full day

Full Demo Rehearsal + Pitch Prep

Run demo 5 times from start to finish. Open two browser windows side by side. Practice judge Q&A answers (see Section 12). Fix any bugs. Record 2-minute

Exactly What You Show Judges (90 Seconds)

1

Open Two Windows Side by Side

Left window: localhost:3000/car — the Hyundai in-cabin dashboard. Right window: localhost:3000/kiosk — the physical charging station. Say: "On the left is what the Hyundai Creta EV driver sees on their car screen. On the right is what is

2

Point to the Battery Level

The dashboard shows battery at 18% and range at 41 km. Say: "Our system detects the car's battery is critically low. It has already queried our AI model and found the best available charger 1.8 km away in Gachibowli."

3

Show the AI Congestion Badge

Click the green pin on the map. Show the station popup: 60 kW DC Fast · CCS2 · AI Demand: LOW · Predicted Queue: 0 minutes. Say: "Our Stanford ROLAND GNN — trained on 36 months of real Telangana data — predicts zero congestion for the

4

Click Reserve 30-Min Exclusive Slot

Click the button. Watch the right window instantly change from GREEN to BLUE. Say: "The moment the driver taps Reserve, our OCPP engine sends a lock command. Watch the station screen change from Available to Reserved — right now, no one

5

Show the PIN on the Car Screen

The dashboard now shows: "Slot Reserved! Your PIN: 4 8 2 9 1 0 | Time remaining: 29:58". Say: "The driver gets a one-time 6-digit PIN shown on the car screen. Nobody else knows this PIN."

6

Try the Wrong PIN — See Lockout

On the kiosk keypad, type 111111. Hit Enter. The kiosk screen flashes: "Access Denied. Contactor remains locked. 0 kW." Say: "This proves the physical lock is real. Even if someone arrives at the station, they cannot steal this charger

7

Type the Correct PIN — Start Charging!

Type 482910 on the kiosk keypad. Watch the screen go: "Identity Verified! Relay Closed! Charging at 60 kW." The car dashboard starts showing a live battery graph climbing from 18%. Say: "The relay closes, 60 kW of DC power flows into

Why This Wins the Ideathon

HEAD-TO-HEAD: WHAT OTHER TEAMS BRING vs SMARTRESERVE

OTHER TEAMS	HYUNDAI SmartReserve
Figma/PowerPoint mockup with fake screenshots	Fully interactive, working live prototype judges can touch
Idea for a mobile app to find chargers	In-cabin experience that lives inside the car — exactly what c
Generic AI buzzwords with no real model	Real trained GNN model with 78% R2 accuracy on official govern
No understanding of charging protocols	OCPP 1.6 protocol implementation — same standard used by Tata
Theoretical reservation concept	Live PIN-based physical lockout demo that judges can physicall

WHY HYUNDAI SPECIFICALLY LOVES THIS

MAKES MONEY No-show fees from reservation system create a new revenue stream. Hyundai takes platform fee from every reservation. Idle fees reduce hogging at	SELLS MORE EVs Range anxiety is the #1 reason Indian consumers do not buy EVs. This system eliminates that fear. More confident buyers = more Creta EV and Ioniq 5
USES EXISTING ASSETS Integrates with myHyundai app and BlueLink telematics that already exist. No new hardware in the car is needed — pure software update.	VALUABLE DATA Anonymous heat maps of where and when people charge help Hyundai and CPO partners decide where to install next fast chargers in India.

Questions Judges Will Ask & Your Answers

Q1

"How does the physical charger know to lock? You cannot install software on real chargers."

YOUR ANSWER:

Every modern DC fast charger in India already runs OCPP 1.6 or 2.0 firmware by law — required by the Ministry of Power's 2024 BEE guidelines. Our backend acts as the Central System (CSMS) and sends standard ReserveNow commands over WebSocket to the

Q2

"What if the CPO (Tata Power, ChargeZone) refuses to give you API access?"

YOUR ANSWER:

That is why Hyundai is the right partner for this. Hyundai already has signed MoUs with ChargeZone, Statiq, and Tata Power for their dealership fast charging network. Hyundai can negotiate API access as part of the partnership terms. For the prototype,

Q3

"Your GNN has 78% accuracy — what about the other 22%? Can we trust it?"

YOUR ANSWER:

We always display the prediction as a range with a confidence label, not a hard promise. The UI shows "Predicted Queue: 0–5 minutes (High Confidence)" or "20–40 minutes (Low Confidence)." This is honest with the driver. We also show the timestamp

Q4

"This is just an app. Why is this an in-vehicle experience?"

YOUR ANSWER:

The car itself detects the battery level and triggers the system automatically — the driver does not have to open any app. The interface runs on the factory-installed AVNT infotainment touchscreen using Hyundai's ccNC connected car platform. The driver uses

Q5

"Is this even legal — deducting money without completing a transaction?"

YOUR ANSWER:

We use UPI AutoPay mandates or card pre-authorization holds — not actual deductions. The money only moves on a confirmed no-show, exactly like hotel booking cancellation fees which are fully legal under RBI guidelines. We would work with Razorpay or PayU

Your Ideathon Submission Outline

- 1

Cover
Hyundai SmartReserve · Category 2: AI-Powered & Digital In-Vehicle Experience ·
- 2

The Problem
4 pain points with real stats: No live status, phone while driving, no reservati
- 3

The Solution
One sentence + the system flow diagram (Car → Cloud → Charger)
- 4

The AI Brain
GNN architecture, 78% accuracy, 934 real stations, 36 months of data
- 5

The Lock Mechanism
OCPP 1.6 ReserveNow flow, charger state diagram, PIN lockout visual
- 6

The In-Vehicle Experience
Screenshot of car dashboard UI — show how the driver never touches their phone
- 7

Live Demo
Run the live prototype: Reserve → Kiosk locks → Wrong PIN → Correct PIN → Chargi
- 8

Why Hyundai
Existing CPO partnerships, myHyundai/BlueLink integration, dealership charger mo
- 9

Business Value
Revenue from no-show fees, more EV sales, data for CPO expansion decisions
- 10

Roadmap
Phase 1: Demo (done) → Phase 2: Pilot 5 dealerships in Hyderabad → Phase 3: All

THE WINNING FORMULA		
Real Data (BEE + Parivahan) + Real AI (GNN 78% R2) + Real Protocol (OCPP 1.6) + Live Interactive Prototype = The team that built the most complete, credible, and real		
PRELIMINARY WINNER	FINAL ROUND	PRIZE
October 30, 2026	December 16, 2026 · Delhi	Rs.1 Lakh finalist + Rs.3 Lakh Grand Prize

Hyundai SmartReserve

Category 2 · Track B · Technology & Prototype · Hyundai CoE Mobility Innovation Ideathon 2026